MIL-E-27675B(USAF)
AMENDMENT 2
21 MARCH 83
SUPERSEDING
AMENDMENT 1
9 April 1973

MILITARY SPECIFICATION

ENGINE PRESSURE RATIO SYSTEM, TURBOFAN ENGINE A/A24J-13

This amendment forms a part of Military Specification MIL-E-27675B (USAF) dated 10 August 1964.

PAGE 1

2.1: Under "SPECIFICATIONS, Federal". Delete "QQ-P-416" and add:

"PPP-B-601 - Boxes, Wood, Cleated Plywood
PPP-B-636 - Box, Shipping, Fiberboard"

2.1: Under "SPECIFICATIONS, Military", delete "MIL-E-5272", "MIL-I-6181", "MIL-P-7936", "MIL-M-26512" and "MIL-D-70327".

2.1: Under "SPECIFICATIONS, Military", add:

"DOD-D-1000 - Drawing, Engineering and Associated List
MIL-C-83488 - Coating, Aluminum, Ion Vapor Deposited"

Add boxed paragraph to bottom center of page.

"Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Oklahoma City Air Logistics Center/MMEDO, Tinker AFB OK 73145 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter."

PAGE 2

2.1: Under "STANDARDS, Military", delete "MS33586" and "AND 10050".

2.1: Under "STANDARDS, Military", add:

"MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control Electromagnetic Interference
MIL-STD-470 - Maintainability Program Requirements (For Systems and Equipments)
MIL-STD-471 - Maintainability Demonstration
MIL-STD-794 - Part and Equipment, Procedures for Packaging and Packing of
MIL-STD-810 - Environmental Test Methods
MIL-STD-889 - Dissimilar Metals
MS33649 - Bosses, Fluid Connection-Internalstraight Thread

FSC 6620

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(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer)."

3.1: Delete and substitute the following:

"3.1 First article. When specified, (see 6.4) the contractor shall furnish sample unit(s) for first article inspection and approval (see 4.2).

PAGE 3

3.4.2.1: Delete reference to "MS33586" and substitute "MIL-STD-889".

PAGE 4

3.5.2: Delete the last sentence.

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Figure 1: Delete reference to "AND10050" and substitute "MS33649-4".

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3.8: Delete reference to "MIL-D-70327" and substitute "DOD-D-1000".

3.9: Delete reference to "MIL-I-6181" and substitute "MIL-STD-461".

3.10.1: Delete paragraph and substitute the following:

"3.10.1 Steel parts. Steel parts, other than corrosion-resistant steel and not in moving contact, shall be coated with ion vapor deposited aluminum, where practicable, in accordance with MIL-C-83488, type I or II as applicable, and of a class that is adequate to achieve the degree of protection required. Other protective coating, in lieu of MIL-C-83488, may be used if demonstrated to be satisfactory and approved by the cognizant engineering activity. Cadmium plating must be avoided when satisfactory alternative processes can be used. Stainless steel parts shall be passivated.

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Add new paragraph 3.14 as follows:

"3.14 Reclaimed materials. The use of reclaimed materials shall be encouraged to the maximum extent possible without jeopardizing the intended use of the item."

4.1: Delete and substitute the following:

"4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise

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specified in the contract order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements."

4.2,a: Delete "Qualification" and substitute "First article".

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4.4 through 4.4.3.1: Delete "Qualification" where referenced and substitute "First article".

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4.6.1.15: Delete paragraph and substitute the following:

"4.6.1.15 Vibration. The transmitter shall be tested within the range of frequencies and vibratory accelerations specified in MIL-STD-810, Method 514.2, Procedure I, Part 1, with vibration isolators, curve D with following changes: (1) Starting amplitude of 0.080 inch; (2) first break point at 10 Hz; (3) second break point at 15 Hz with amplitude of 0.036 inch; and (4) third break point at 75 Hz. Function of the transmitter shall be checked concurrently with the operation of resonance frequency search. This test shall be conducted at room temperature.

4.6.1.15.1: Delete paragraph and substitute the following:

"4.6.1.15.1 Resonant frequencies. When resonant frequencies are not apparent within the specified frequency range, the transmitter shall be vibrated for a two-hour period with vibration isolators at a frequency of 55 Hz and an amplitude of 0.060 inch on each of three mutually perpendicular axes."

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4.6.1.15.1.1: Delete paragraph entirely.

4.6.1.16: Delete paragraph and substitute:

"4.6.1.16 Humidity. Except when hermetically sealed, the transmitter shall be subjected to the humidity test in accordance with MIL-STD-810, Method 507.1, Procedure I., with the following changes: (1) Step 2: Raise chamber temperature to 71°C (160°F); (2) Step 4: Maintain 95 + 5 - 3 percent humidity; (3) Step 6: Change 85% to 95 + 5 - 3%."

4.6.1.17: Delete and substitute the following:

"4.6.1.17 Fungus. Except when completely hermetically sealed, the transmitter shall be subjected to the fungus test in accordance with MIL-STD-810, Method 508.1, Procedure I, for a period of 14 days."

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4.6.1.18: Delete the first sentence and substitute "4.6.1.18 Salt fog. The transmitter shall be subjected to the salt fog test in accordance with MIL-STD-810, Method 509.1, Procedure I."

4.6.1.19: Delete and substitute the following:

"4.6.1.19 Dust (fine sand). The transmitter shall be subjected to dust test in accordance with MIL-STD-810, Method 510.1, Procedure I, with the following changes: (1) Step 1: Adjust the air velocity to 300 ± 200 feet per minute; (2) Step 2: Maintain dust feed and air velocity at 300 ± 200 feet per minute. Raise the internal chamber air temperature to 71°C (160°F). Hold these conditions for 6 hours; and (3) Step 3: Delete. After dust test, subject transmitter to short scale error at room temperature test specified in 4.6.1.3.1. The tolerances shall not exceed those specified in Table V."

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4.6.2.5: Delete first sentence only and substitute the following:

"4.6.2.5 Vibration. The indicator shall be subjected to the vibration test in accordance with MIL-STD-810, Method 514.2, Procedure I, Part 1, without vibration isolators with the following changes: (1) Use curve B; (2) starting amplitude of 0.010 inch; and (3) break point at 60 Hz."

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4.6.2.17: Delete first sentence an substitute:

"4.6.2.17 Salt fog. The indicator shall be subjected to a salt fog test in accordance with MIL-STD-810, Method 509.1, Procedure I."

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4.6.2.19: Delete reference to "MIL-I-6181" and substitute "MIL-STD-461".

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Add the following new paragraph:

"4.7 Inspection and test. Tests of methods of preservation and packaging shall be accomplished in accordance with Section 4 of MIL-P-116, to insure compliance with Section 5 of this specification."

Section 5: Delete and substitute the following:

"5. PACKAGING

5.1 Preservation-packaging. Preservation-packaging shall be Level A or C as specified.

5.1.1 Level A.

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5.1.1.1 Cleaning. Components of the system shall be cleaned in accordance with process C-1 of MIL-P-116."

5.1.1.2 Drying. Components of the system shall be dried in accordance with process D-4 of MIL-P-116.

5.1.1.3 Preservation application. Not applicable.

5.1.1.4 Unit packaging. Unless otherwise specified by the procuring activity, components of the system shall be packaged in quantity unit packs of one each in accordance with method IA of MIL-P-116. Each component shall be overboxed in PPP-B-636 carton. Carton shall be large enough to allow for application of sufficient cushioning material between container and barrier, of a type, density, and thickness to insure shock transmission does not exceed peak values in G's established for the components when completed packs are subjected to the rough handling drop tests of MIL-P-116.

5.1.2 Level C. Components shall be clean, dry, and individually packaged in a manner that will afford adequate protection against corrosion, deterioration, and physical damage during shipment from supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B, or C as specified.

5.2.1 Level A. Components packaged as specified in 5.1.1 shall be packed in shipping containers conforming to PPP-B-601, styles A and B, Class Overseas, unless otherwise specified by the procuring activity. Insofar as practical, exterior shipping containers shall be of uniform shape, size and of minimum tare and cube, consistent with the protection required.

5.2.2 Level B. Components packaged as specified in 5.1.1 shall be packed in weather resistant class exterior shipping containers conforming to PPP-B-636. Other requirements as specified in 5.2.1 above are applicable.

5.2.3 Level C. Packing shall be applied which affords adequate protection during domestic shipment from the supply source to the first domestic receiving activity for immediate use. This level shall conform to applicable carrier rules and regulations and may be the contractor's commercial practice provided the latter meets the requirements of this level.

5.3 Marking. In addition to any special marking required by the contract or order, each unit package and shipping container shall be marked in accordance with MIL-STD-129."

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6.4: Delete paragraph and substitute the following:

"6.4 First article. When a first article is required, the item will be inspected and tested in accordance with Section 4 of this document. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, test and approval of the first article."

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NOTE: Asterisks were not used in this amendment to denote changes with respect to the previous issue due to the extensiveness of the changes.

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